Weekly Business Review
Olist E-commerce Marketplace

Objective: Evaluate delivery reliability, customer experience, order value, product-category performance, and repeat purchasing using the Olist Brazilian E-Commerce dataset.

1. Executive Summary

The analysis mart contains 99,441 rows and 99,441 distinct order IDs. The on-time delivery rate is 91.89%, the late-delivery rate is 8.11%, the average review score is 4.09/5, and average order value is 159.33 in the dataset's currency, including product price and freight.

The North Star metric is on-time delivery rate because it directly measures whether delivered orders meet the estimated delivery date. Review scores provide a customer-experience measure alongside it.

2. Pipeline and Methodology

Raw ingestion: CSV files are loaded into PostgreSQL's raw schema. **Staging:** Relevant fields and data types are prepared. **Analysis mart:** Item, payment, and review data are aggregated before joining to orders to avoid multiplying order-level records. **Analysis:** SQL calculates monthly value, state-level categories, rolling order counts, repeat purchasing, and delivery/review measures.

The pipeline log confirms successful execution and verifies that the mart has 99,441 rows and 99,441 unique orders.

3. Data Quality Results

The automated pipeline ran nine checks. Six critical checks returned zero failures. Three additional checks identified exceptions that were flagged for investigation rather than silently discarded.

Check	Failed rows	Handling
Missing order IDs	0	Passed
Duplicate order IDs	0	Passed
Invalid delivery dates	0	Passed
Invalid item values	0	Passed
Missing customer records	0	Passed
Invalid review scores	0	Passed
Payment mismatches	267	Flag for investigation
Orders without items	775	Flag; review status
Orders without payments	1	Flag for investigation

The 267 payment mismatches indicate differences greater than 0.02 between aggregated item-plus-freight totals and payment totals. The 775 orders without items are largely associated with unavailable or canceled statuses, but all exceptions should be reviewed. One delivered order has no matching payment record. No exception should be assumed fraudulent or deleted without investigation.

4. Supporting Business Metrics

Metric	Result	Definition
On-time delivery	91.89%	Delivered on/before estimate
Late delivery	8.11%	Delivered after estimate
Average review	4.09/5	Mean review score
Average order value	159.33	Mean item price + freight per order
Repeat purchase rate	3.12%	Unique customers with >1 order / unique customers
Repeat-customer order share	6.38%	Orders from repeat customers / all orders

5. Commercial Performance

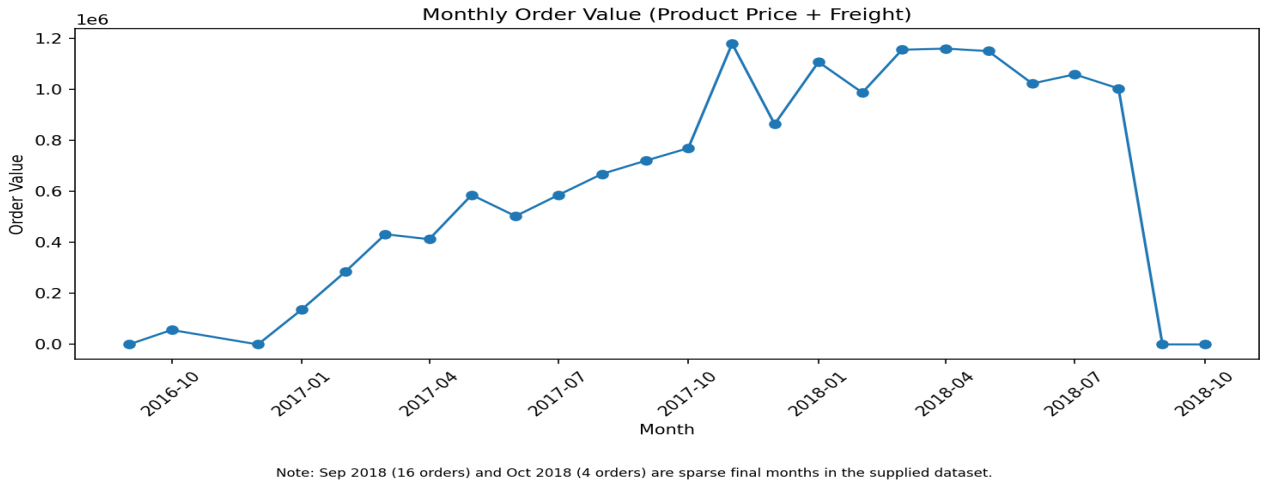


Figure 1. Monthly order value and month-over-month growth.

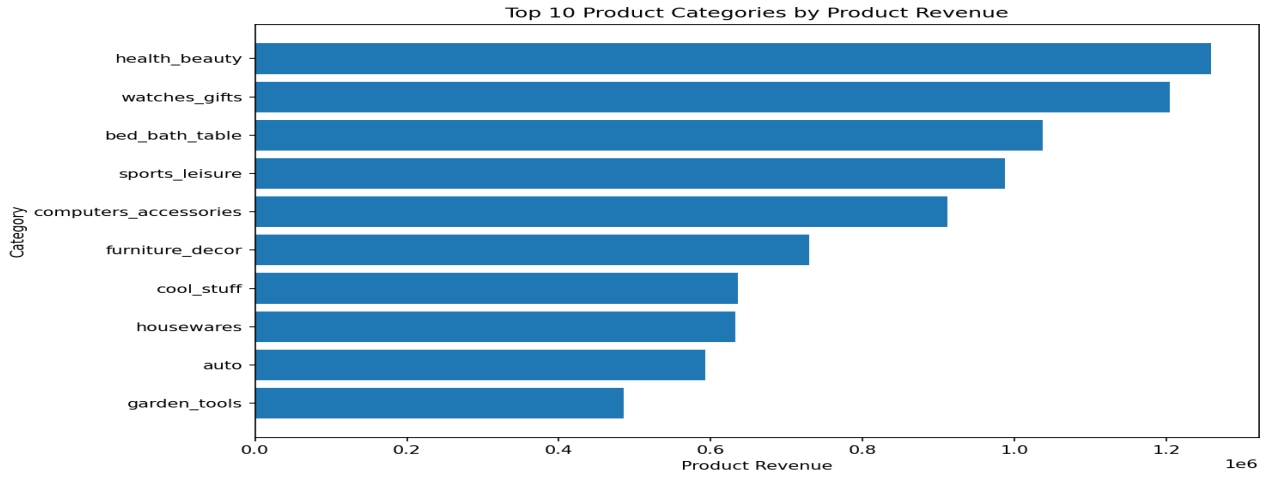


Figure 2. Top product categories by state.

Interpret monthly movements together with data coverage and order counts. Category rankings identify where product demand is concentrated and may help guide state-specific assortment decisions.

6. Orders and Customer Experience

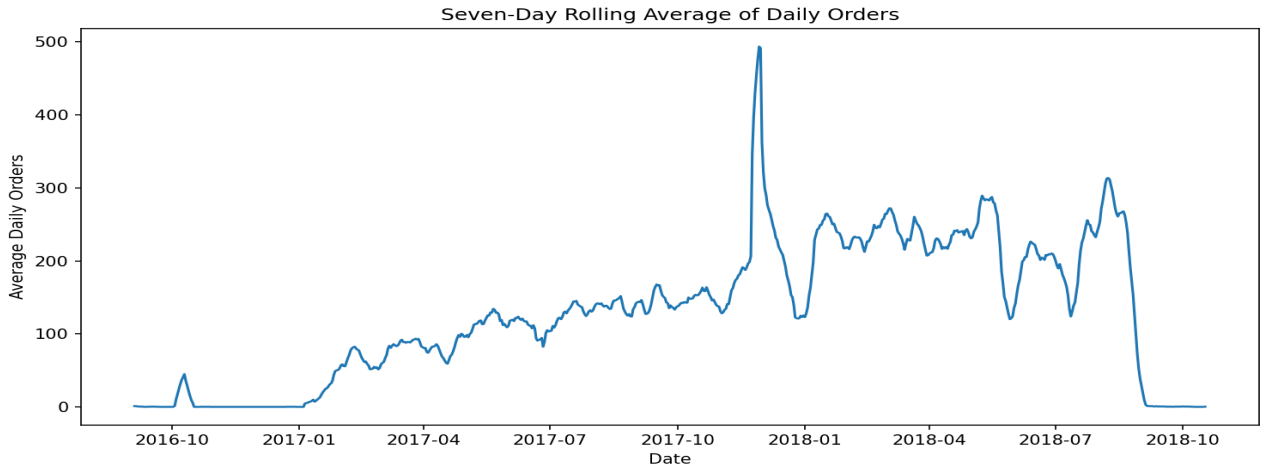


Figure 3. Seven-day rolling average of daily orders.

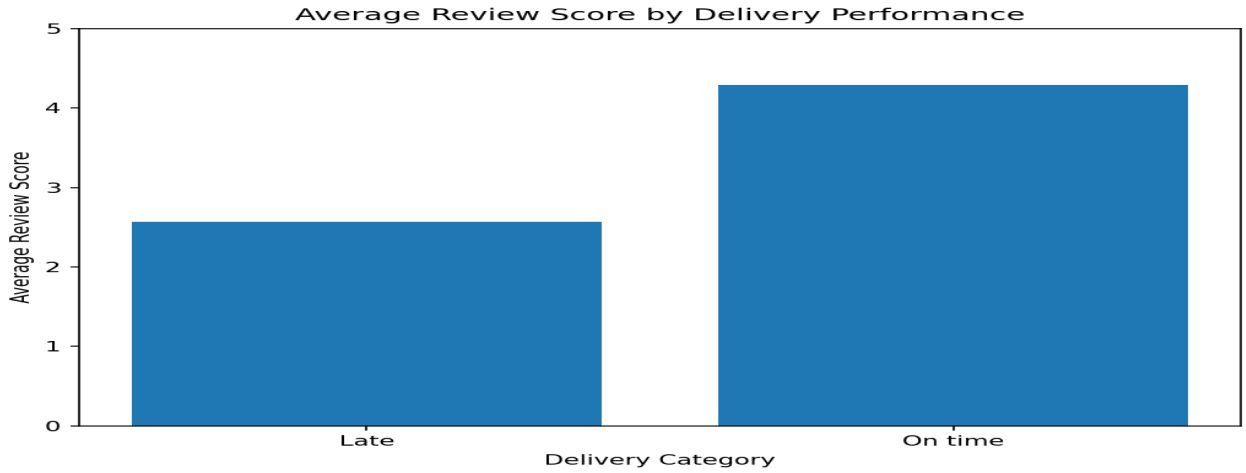


Figure 4. Review scores grouped by delivery performance.

Late delivery is associated with a lower average review score in this dataset, but this relationship alone does not establish causation. Delivery and review metrics should be monitored together.

7. Recommendations for the VP

1. Reduce late deliveries. Investigate late-delivery rates by customer state, seller, product category, and delivery period. Prioritize the segments contributing most to late orders.

2. Monitor customer experience. Review on-time delivery and average review scores weekly. Compare changes over time before concluding that one measure caused the other.

3. Improve retention. Segment customers by order history and examine repeat purchasing before testing targeted retention initiatives.

4. Resolve data exceptions. Investigate payment mismatches, orders without items, and the delivered order without a payment record. Document any corrections and rerun the quality checks.

Metric Definitions

On-time delivery rate: delivered orders with actual delivery date on or before estimated delivery date divided by orders with known delivery outcomes. **Late-delivery rate:** delivered orders after the estimate divided by orders with known delivery outcomes.

Average order value: mean of product price plus freight per order. It is not necessarily accounting revenue. **Repeat purchase rate:** unique customers with more than one order divided by all unique customers. **Repeat-customer order share:** orders placed by repeat customers divided by all orders.

Conclusion

The current results establish a baseline for weekly business reviews. The pipeline ran successfully and the mart passed its row-uniqueness verification. Data-quality exceptions and unusual monthly movements should be investigated before the results are used for operational or financial decisions.

Dataset: Olist Brazilian E-Commerce dataset. Analysis tools: Python, PostgreSQL, SQL, pandas, and matplotlib.